$= (xyz + x'yz) + (xy'z + xyz' + xyz)$ (Theorem 4(a))
$= (x \cdot yz + x' \cdot yz) + (xy'z + xyz' + xyz)$ (Theorem 4(b))
$= (yz \cdot x + yz \cdot x') + (xy'z + xyz' + xyz)$ (Postulate 3(b))
$= (yz \cdot (x + x')) + (xy'z + xyz' + xyz)$ (Postulate 4(a))
$= (yz \cdot 1) + (xy'z + xyz' + xyz)$ (Postulate 5(a))
$= yz + (xy'z + xyz' + xyz)$ (Postulate 2(b))
$= yz + (xy'z + xyz' + (xyz + xyz))$ (Theorem 1(a))
$= yz + ((xy'z + xyz' + xyz) + xyz)$ (Theorem 4(a))
$= yz + (xyz + (xy'z + xyz' + xyz))$ (Postulate 3(a))
$= yz + ((xyz + xy'z) + (xyz + xyz))$ (Theorem 4(a))
$= yz + ((yx \cdot z + y'x \cdot z) + (xyz + xyz))$ (Postulate 3(b))
$= yz + ((y \cdot xz + y' \cdot xz) + (xyz + xyz))$ (Theorem 4(b))
$= yz + ((xz \cdot y + xz \cdot y') + (xy \cdot z + xyz))$ (Postulate 3(b))
$= yz + ((xz \cdot (y + y')) + xy \cdot (z + z))$ (Postulate 4(a))
$= yz + ((xz \cdot (y + y')) + xy \cdot (z + z'))$ (Postulate 3(a))
$= yz + ((xz \cdot 1) + xy \cdot 1))$ (Postulate 5(a))
$= yz + (xz + xy)$ (Postulate 2(b))
$= yz + xz + xy$ (Theorem 4(a))

2-21: Verify the truth table for the 3-input exclusive-OR gate listed in Fig. 2-8(c). List all eight combinations of x, y and z; evaluate $A = x \oplus y$; then evaluate $F = A \oplus z = x \oplus y \oplus z$.

x	y	z	$A = x \oplus y$	$F = A \oplus z$
0	0	0	0	0
0	0	1	0	1
0	1	0	1	1
0	1	1	1	0
1	0	0	1	1
1	0	1	1	0
1	1	0	0	0
1	1	1	0	1